

The Execution-Certainty Wedge

How Measurement Choices Rank Housing-Access Instruments

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Abstract

Sellers can prefer cash offers even when financed buyers value the home more. This paper compares housing-access instruments in an agent-based market. Their ranking depends on measurement. An efficiency benchmark computed over buyers still waiting moves with the policy and reverses the sign of the contention duty's effect for offer discounts between about 5% and 12%. Completed-cohort access gains are 1.4 to 2.5 times the full-run gains, and the two measures can rank instruments differently. A constant charge preserves more value than the duty at common full-run access targets; the duty preserves slightly more at the ten-point cohort target. A quota that resets each period cannot reach small access targets at low institutional participation, while carrying allowances forward removes that limit. At New York City company-purchase proxy values, removing the cash discount entirely lowers the share of owner-occupiers who leave without buying only from 9.8% to 9.5%, so the certainty wedge accounts for little of the model's access shortfall there. These comparisons retain illustrative market conditions; transaction-price gaps and company status provide incomplete evidence on execution certainty and institutional participation.

Keywords: execution certainty; housing access; policy evaluation; efficiency measurement; agent-based model; transaction taxes

JEL classification: D47; D61; D63; H23; R31

1. Introduction

A seller choosing between two offers for the same home may take the lower one because it is more likely to close. Cash offers usually are. When the buyers who can pay cash are mainly investors and the buyers who need a mortgage are mainly households who would live in the home, the seller's preference can decide who gets housing. In May 2026 New York legislators considered a 1% tax on all-cash residential purchases of \$1 million or more. It was reported as likely to be dropped a week later and was left out of the budget passed on 27 May (Bloomberg, 14, 21 and 27 May 2026). The proposal raises a design question that does not depend on its fate: if a charge on one class of buyer is meant to change who wins, how should its effects be measured?

The preference for certainty is not in itself a distortion. A failed sale costs the seller time, carrying costs and a relisting, and discounting a riskier offer can maximise expected surplus. Reher and Valkanov (2024) estimate a national mortgage-cash premium larger than their model of actuarial risk explains. Han and Hong (2024) find a smaller discount in Los Angeles and show that closing and relisting risks together can account for it. The companion New York City study (Loschi 2026a) finds a financing-related price gap of about 9 log points for houses and close to zero for condominiums. Within houses the gap is small when the cash buyer is a company, and it falls by almost half when pairs with a same-surname deed, a marker of possible transfers between relatives, are excluded. Much of the observed discount therefore reflects who the cash buyers are, not what closing certainty is worth.

That evidence limits what a model of the certainty wedge can claim about New York. This paper asks a narrower question. Suppose certainty does decide some allocations. Which instruments change them, at what cost, and how would an evaluator know? The answer depends on measurement choices that are easy to make without noticing.

The model is a stylised market with owner-occupiers, small investors and institutional buyers. Sellers rank offers by valuation times a certainty factor, and institutional offers are the most certain. Four instruments are compared with an unconstrained market: a cap on acquisitions per legal entity; a work-conserving fair share that limits institutional purchases only while owner-occupiers are waiting; a constant proportional charge on institutional offers; and a duty whose rate rises with a local contention signal, the share of owner-occupiers

who leave without buying. A fixed-offer result, Proposition 1, shows which acquisitions a proportional charge displaces. The aggregate comparison is computational.

Four findings concern measurement. First, an efficiency benchmark computed over the buyers present at each sale depends on the policy, because the policy decides who is still waiting. In this model it disagrees in sign with a fixed benchmark whenever the offer discount lies between about 5.4% and 11.7%, and a two-period example shows why: a policy that serves a high-value buyer at once removes that buyer from the queue on which the benchmark is computed. Second, the usual access statistic, exits without purchase over the whole run, counts buyers still searching at the end as neither served nor unserved. Following a cohort until every member has bought or left shows access gains 1.4 to 2.5 times larger, and the two measures rank the instruments differently. Third, the comparison between the contention duty and a constant charge depends on which access population the instruments are tuned to. The constant charge preserves more value at common full-run targets, and the duty keeps slightly more at the ten-point cohort target. Fourth, the fair-share quota's failure to reach small targets at low institutional participation comes from its period-reset accounting. Allowances that carry forward reach intermediate targets.

Two results concern the instruments themselves. At the baseline the duty lowers the share of owner-occupiers who leave without buying from 41.9% to 36.6% with almost no change in total realised valuation, while the cap lowers it to 8.7% at an 11.3% loss. Buyer-class weights can reverse that ranking: with owner-occupier valuations weighted by about 1.66 or more, the cap scores highest of the three original instruments.

A new sensitivity exercise uses New York City company-purchase proxies to set the offer discount and approximate institutional participation, with both full-run and completed-cohort access reported.

Section 2 places the paper in the literature, Section 3 sets out the model and the measures, and Section 4 the seller's problem and the selection result. The comparisons in Section 5 are organised by the measurement choice that changes them. Section 6 compares the break-even offer discount with New York City evidence and runs the model at proxy values. Section 7 sets out what the model leaves out, Section 8 what each instrument would need in practice, and Section 9 concludes. Appendix A records how the results can be reproduced. The extended version of this paper (Loschi 2026b) contains the full robustness exercises, a longer discussion of implementation and objections, and the full replication record.

2. Related work

The paper sits between housing finance, market design and public finance. Reher and Valkanov (2024) and Han and Hong (2024) supply the empirical motivation and disagree on how much of the cash premium is excess over risk; that disagreement is why this paper sweeps the offer discount instead of transporting one estimate. Che and Gale (1998) show that auctions with budget-constrained bidders can allocate inefficiently. The mechanism here runs through the seller's preference for certainty instead, and it operates even when the buyer's willingness to pay is unconstrained. Budish, Cramton and Shim (2015) study how market design can reward an advantage other than valuation, in their case speed. Completion certainty differs because it can save real costs.

The policy question is a second-best one in the sense of Lipsey and Lancaster (1956): a charge in an already distorted market can raise or lower welfare, and nothing general follows. Comparing a quantity rule (a cap or quota) with a price rule (a charge) follows Weitzman (1974), and the carry-forward quota borrows the logic of banking allowances analysed in Weitzman (2018). The treatment of distribution follows the practice of stating welfare weights explicitly (Atkinson 1970) and of reporting the trade-off between efficiency and fairness instead of assuming one objective (Bertsimas, Farias and Trichakis 2012). The method is agent-based in the sense of Tesfatsion (2006). Bidding is a fixed rule, not a learning or equilibrium strategy, which is the model's largest limitation.

The measurement findings have close relatives outside housing. A benchmark that moves with the treatment is a familiar problem in queueing systems, and a waiting-time statistic computed only over served requests hides the ones that were dropped. The contribution here is to show how much these choices matter for ranking housing-access instruments.

3. The model

3.1 Market

The simulation is a discrete-time market with five districts over 120 periods, calibrated to be illustrative, not empirical. Each district lists a Poisson-distributed number of units averaging 20 per period. Three classes of buyer bid:

- **Owner-occupiers** arrive at 17 per district per period, draw valuations from a lognormal distribution centred near 400,000 ($\sigma = 0.34$), and leave unserved after 36 periods.
- **Small investors** arrive at 4 per district per period, centred near 372,000 ($\sigma = 0.30$), with 24 periods of patience.
- **Institutional buyers** are three standing entities that together can place 42 bids per district per period, centred near 366,000 ($\sigma = 0.18$).

Sellers rank offers by effective value, the bid multiplied by a certainty factor: 1.00 for institutional cash offers, 0.95 for small investors and 0.88 for financed owner-occupiers. The owner-occupier factor is a 12% offer discount, $w = 1 - c$. It is an illustrative parameter, and Section 5.2 sweeps it. A financed price premium p over the cash price corresponds to $w = p/(1 + p)$, and a log-price gap β to $w = 1 - \exp(-\beta)$. Units go to the highest effective offer in the eligible pool. No bargaining equilibrium is solved.

3.2 Instruments

Unconstrained. Sellers accept the highest effective offer.

Static per-entity cap. Each legal entity may acquire at most 50 units per district over the run.

Work-conserving fair share. Institutional buyers may take at most 30% of a district's listings in a period while owner-occupiers are waiting. When none is waiting, the limit does not apply.

Contention-triggered duty. Institutional effective offers are multiplied by $(1 - \tau_{\max} \cdot s)$, with $\tau_{\max} = 0.22$. The signal s is measured on exits from the local owner-occupier cohort. Over a trailing window of six periods, let U be the number who left without buying and V the number who bought; then $s = U/(U + V)$, and $s = 0$ when the window is empty. Buyers still searching count in neither term. The signal counts exits because a bid-based version, the share of owner-occupier bids that lose, saturates near one in a market with hundreds of waiting buyers and twenty units per period, and would define a different instrument. Across regimes the realised mean of s runs from 0.078 under the cap to 0.406 without intervention; under the duty it is 0.341. The multiplier is a reduced-form offer haircut. It is not a statutory tax rate, and no revenue is computed.

Constant charge. Institutional effective offers are multiplied by $(1 - h)$, whatever the contention. It needs no registry of entries and exits, although classifying eligible buyers and enforcing the charge still require administration.

Combined. The fair share and the duty together.

3.3 Measures

The primary outcome is the **realised-value index**: the sum of successful buyers' valuations, normalised to the unconstrained regime under matched seeds. It measures realised willingness to pay, not social welfare. It omits renters, the use of any revenue, administrative costs and differences in the marginal utility of income.

Two efficiency ratios put that sum over a benchmark. **Oracle efficiency** divides by the sum of the L highest valuations among all bidders generated in the run, where L is the number of units listed. With full search and exogenous listings, matched seeds give every policy the same denominator. **Contemporaneous efficiency** divides by the sum, over listings, of the highest valuation among the bidders present at that listing. That denominator depends on who is still waiting, and so on the policy.

Access is measured by the **starvation rate**: owner-occupiers who leave without buying, divided by all owner-occupier arrivals during the run. Buyers still searching at the horizon stay in the denominator. Section 5.3 contrasts this with a completed-cohort measure. Results are means over twelve independent replications unless stated otherwise.

4. The seller's problem and which bids a charge displaces

4.1 How much discount risk justifies

A seller holds a unit worth P . A cash offer C closes at horizon τ_C . A financed offer F closes with probability $1 - q$ at $\tau_F > \tau_C$; if it fails, the seller incurs a relisting loss δ and eventually sells at P . With carrying cost c per period and risk neutrality, the seller is indifferent when

$$(1 - q)(F - c\tau_F) + q(P - \delta - c\tau_F) = C - c\tau_C. \quad (1)$$

Setting $P = C$, $d = \delta/C$ and $\kappa = c(\tau_F - \tau_C)/C$, the justified discount on the financed price, $\pi^* = (F - C)/F$, is

$$\pi^* = (qd + \kappa) / (1 - q + qd + \kappa). \quad (2)$$

It depends on the failure probability, the cost of a failed sale and the difference in closing times, and it is positive even when no sale fails (at $q = 0$ it equals $\kappa/(1 + \kappa)$). Only a discount beyond π^* is a candidate distortion, and neither a price gap nor a mortgage denial rate identifies π^* without local evidence on accepted offers that fail.

4.2 Which bids a proportional charge displaces

Consider one listing with fixed offers and eligibility. A targeted buyer j has valuation v_j and certainty factor c_j ; an untargeted buyer i has v_i and c_i . Moving the unit from j to i changes realised valuation by $-G$, where $G = v_j - v_i$. This local measure ignores real completion costs.

Proposition 1 (fixed-offer selection). *Let $a = v_j c_j$ be the highest effective offer among targeted buyers and $b = v_i c_i$ the highest among untargeted buyers. Suppose $a \geq b > 0$, ties go to the targeted group, and offers and eligibility are fixed. Multiplying every targeted offer by $(1 - \theta)$, with $0 < \theta < 1$, changes the winning group exactly when*

$$r = 1 - b/a < \theta. \quad (3)$$

If $c_j > c_i$ and $\theta \leq 1 - c_i/c_j$, every such displacement has $v_i > v_j$.

Proof. Scaling all targeted offers by the same factor preserves their ranking, so the best targeted offer after the charge is $(1 - \theta)a$. The untargeted group wins if and only if $b > (1 - \theta)a$, which is (3). A displacement implies $v_i/v_j > (1 - \theta)c_j/c_i$, which is at least one under the stated condition. ■

The charge therefore selects listings by their normalised effective-offer margin. This does not rank its average loss against a cap's. Take two equally likely listings with targeted and untargeted valuations (10,000, 9,700) and (100, 80), and certainty factors 1 and 0.88. A 20% charge changes the winner only at the first listing, at a valuation loss of 300. Excluding the targeted buyer with probability one-half at each listing gives the same expected number of displacements and a conditional average loss of 160. Aggregate comparisons have to be computed.

In the simulation they come out as follows. Removing the current instrument from the candidate pool at each listing and comparing winners gives a mean G of +143,789 for the cap, +22,188 for the fair share and -12,905 for the duty, with 9.0%, 32.8% and 80.4% of displacements replacing the winner with a higher-value buyer. The duty mostly moves units to buyers who value them more; the cap mostly does the opposite. These local comparisons cannot be summed into a policy effect, because a buyer served at one listing might otherwise have bought later.

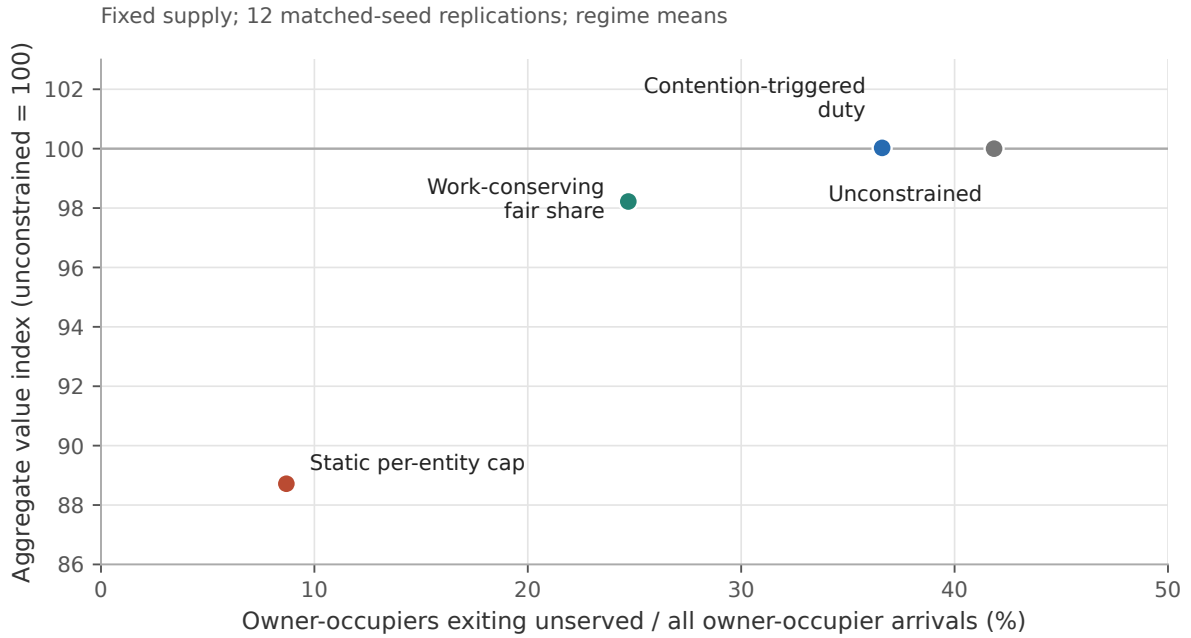
5. Results

5.1 The baseline comparison

Regime	OO share	Inst. share	Starvation	Value index	Oracle eff.	Contemp. eff.
Unconstrained	34.0%	58.1%	41.9%	100.00	99.5%	95.4%
Static per-entity cap	75.8%	6.3%	8.7%	88.72	88.3%	78.5%
Work-conserving fair share	54.5%	32.2%	24.7%	98.22	97.7%	95.2%
Contention-triggered duty	41.1%	49.4%	36.6%	100.03	99.5%	98.1%
Combined (share + duty)	54.7%	32.1%	24.6%	98.20	97.7%	96.3%

Table 1. Allocation regimes at the baseline calibration, means of 12 replications. OO and Inst. share are the shares of units bought by owner-occupiers and institutions. The value index is the sum of winners' valuations relative to the unconstrained regime under matched seeds.

Distribution and aggregate value at the baseline calibration



Value = sum of winners' true valuations. Means shown without Monte Carlo uncertainty bars.

Figure 1. Access and realised value at the baseline, means over 12 replications. Starvation is owner-occupier exits without purchase over all owner-occupier arrivals. The vertical axis is the value index. Points are means, not confidence regions.

The instruments deliver very different amounts of access (Table 1, Figure 1). The duty lowers starvation from 41.9% to 36.6% with a value index of 100.03; its gain of about 0.026 index points has a paired Monte Carlo standard error of 0.0054, so it is precisely estimated and economically negligible. Starvation falls to 24.7% under the fair share, at a 1.8% loss of value, and to 8.7% under the cap, at an 11.3% loss. Combining the fair share and the duty adds almost nothing, because the fair share lowers the contention signal and with it the duty's rate. The unconstrained market reaches 99.5% of the oracle benchmark, so the room for aggregate efficiency gains is small; the differences that are large are distributional.

Whether a larger access gain is worth its cost depends on the welfare weights. Scoring each allocation with $W(\omega) = \omega V_{OO} + \sqrt{\omega} V_{SI} + V_I$, where V is the realised valuation of each class, the fair share beats the duty from $\omega \approx 1.16$ and the cap becomes the best of the three original instruments from $\omega \approx 1.66$. The cap breaks even against no intervention at $\omega \approx 1.35$. These weights are normative scenarios, not estimates.

ω	Static cap	Fair share	Duty	Highest
1.00	88.72	98.22	100.03	duty
1.10	92.32	100.04	100.66	duty
1.20	95.62	101.70	101.24	fair share
1.50	104.08	105.96	102.72	fair share
2.00	114.72	111.32	104.59	cap

Table 2. Class-weighted realised valuation, $W(\omega)$, relative to the unconstrained regime at the same weight; ratio of means over 12 matched replications. $\omega = 1$ recovers the unweighted index. “Highest” compares only the three instruments shown, with fixed supply.

Two constraints from the extended version change the cap’s case in particular. When institutional capital is split across legal entities at no cost, the cap’s effect decays and disappears at 25 entities; beneficial-owner aggregation would be needed to enforce it. When listings respond to seller proceeds with unit elasticity and the seller bears the whole charge, the cap’s value loss rises to 17.3% and its starvation rate to 16.0%, while the duty loses 2.0 index points. The duty stays the least costly in value at every elasticity tested, but it is not costless.

Regime	$\eta = 0$: value / starvation	$\eta = 0.5$	$\eta = 1.0$
Unconstrained	100.00 / 41.9%	100.00 / 41.9%	100.00 / 41.9%
Static per-entity cap	88.73 / 8.8%	85.82 / 12.5%	82.67 / 16.0%
Work-conserving fair share	98.23 / 24.8%	97.49 / 25.2%	97.38 / 25.3%
Contention-triggered duty	100.02 / 36.6%	99.25 / 36.9%	97.97 / 37.1%
Combined	98.20 / 24.7%	97.36 / 25.1%	95.77 / 26.0%

Table 3. Value index and starvation rate when listings respond to seller proceeds with elasticity η , under full seller incidence of the duty; 8 replications. This is a listing response, not a model of construction.

5.2 Which efficiency benchmark

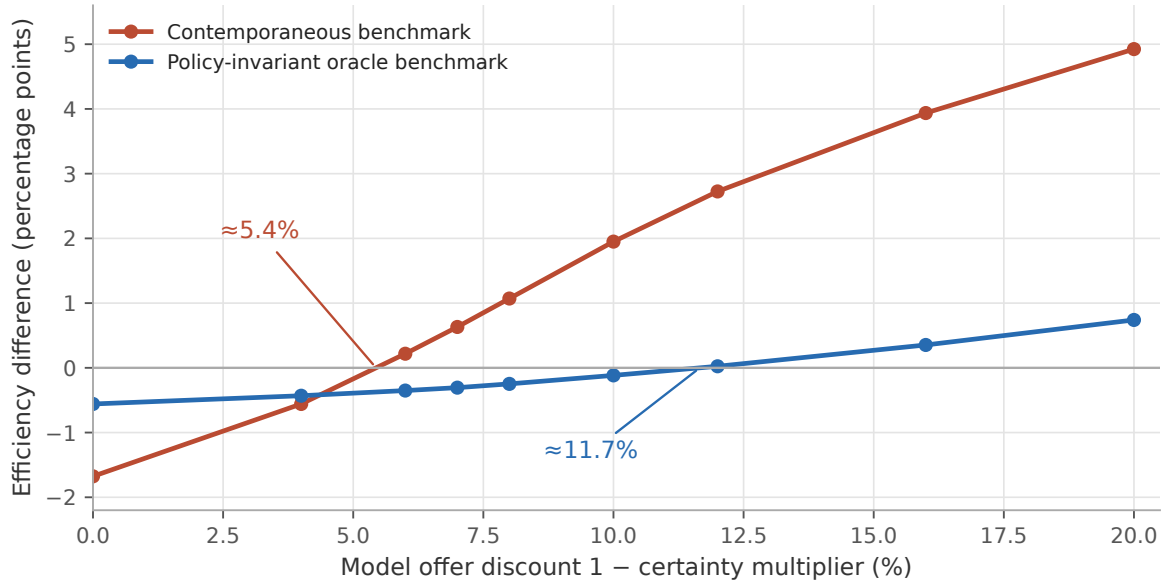
The two efficiency ratios in Table 1 differ systematically. In a market where buyers wait, the policy decides who is waiting. A regime that serves high-value buyers quickly leaves a thinner queue, and a benchmark computed over that queue falls with it. In this model the contemporaneous benchmark averages about 521,000 per unit without intervention, 562,000 under the cap and 506,000 under the duty, a spread of about 11%. Ratios against it are not comparable across regimes.

Offer discount w	Gain, oracle (pp)	Gain, contemporaneous (pp)	Value index, duty
0%	−0.56	−1.68	99.44
4%	−0.43	−0.56	99.57
6%	−0.35	+0.22	99.65
8%	−0.25	+1.07	99.75
10%	−0.12	+1.95	99.88
12%	+0.02	+2.73	100.02
16%	+0.35	+3.94	100.36
20%	+0.74	+4.92	100.75

Table 4. The duty’s efficiency gain over the unconstrained market as the owner-occupier offer discount varies, 8 replications per row. At each row $c_{OO} = 1 - w$ and $c_{SI} = \min(1, c_{OO} + 0.07)$; institutional certainty stays at one.

A moving denominator changes the apparent break-even

Duty minus unconstrained; 8 matched-seed replications; no deal failures



Thresholds use linear interpolation; bootstrap 95% intervals are reported in the text. Gross empirical premiums are n

Figure 2. The sweep of Table 4 on the fixed (oracle) and moving (contemporaneous) benchmarks. Linear interpolation gives crossings near 11.7% and 5.4%. The horizontal axis is the model's offer discount, not an empirical price premium; lines join simulation means.

A two-period example shows the mechanism. One unit is sold in each period. In period 1 an institution with valuation v_I and an owner-occupier with valuation $v_O > v_I$ are present; the institution offers with certainty 1 and leaves if not served, and the owner-occupier's effective offer is $c \cdot v_O < v_I$, but the owner-occupier stays for period 2. In period 2 a late owner-occupier with valuation $v_L < v_I$ arrives. Without intervention the institution wins period 1 and the first owner-occupier wins period 2, for a realised value of $v_I + v_O$. A duty large enough to make $(1 - \theta)v_I < c \cdot v_O$ gives the owner-occupier the unit in period 1; the institution leaves, and the late buyer wins period 2, for $v_O + v_L$. The duty lowers realised value by $v_I - v_L$, and the fixed oracle benchmark, $v_O + v_I$ in both regimes, shows the loss. The moving benchmark counts the highest valuation present at each sale: $2v_O$ without intervention, because the waiting owner-occupier is present at both sales, and $v_O + v_L$ under the duty. Contemporaneous efficiency therefore rises from $(v_I + v_O)/(2v_O)$ to 1. With $v_O = 10$, $v_I = 6$, $v_L = 1$ and $c = 0.5$, it rises from 0.80 to 1.00 while realised value falls from 16 to 11. The policy removed a high-value buyer from the queue, and the benchmark fell with it.

Table 4 and Figure 2 show the consequence in the full model. On the fixed benchmark the duty starts adding value at an offer discount of about 11.7% (95% interval [11.5, 11.8]). On the moving benchmark it appears to add value from about 5.4% [5.4, 5.5]. At the tested discounts of 6%, 8% and 10% the two measures disagree on the sign of the policy's effect: the moving benchmark reports gains of 0.22 to 1.95 points where the fixed one reports losses of 0.12 to 0.35. Choosing the moving benchmark more than doubles the range of discounts over which the duty seems to pay for itself.

The fix is to hold the benchmark independent of the policy. The value index needs only identical arrivals across regimes, and oracle efficiency needs the full arrival stream. A simple diagnostic makes the problem visible: report the benchmark's value per unit alongside each ratio. If it differs across treatment arms, the ratio is counting the treatment twice.

5.3 Which access population

The default instruments differ in strength as well as design, so a fair comparison tunes each to the same access gain. Each instrument was tuned on four simulation seeds to reduce starvation by five or ten percentage points, and the chosen settings were then evaluated on sixteen separate seeds. Near the five-point target the cap, fair

share and duty achieve gains of 5.42, 5.78 and 5.03 points with value indices of 97.46, 99.99 and 100.03. Near ten points the gains are 10.64, 10.14 and 10.05, with indices of 95.73, 99.63 and 99.32. At the stronger target the fair share keeps more value than the duty, so the default ordering between them does not survive tuning.

Target (pp)	Instrument	Setting	Access gain (pp; SE)	Value index (SE)	Constant – duty, value (SE)
5	Constant charge	0.0779	4.97 (0.041)	100.08 (0.005)	+0.051 (0.003)
5	Signal duty	0.2115	5.03 (0.031)	100.03 (0.005)	—
10	Constant charge	0.1466	9.79 (0.034)	99.68 (0.008)	+0.361 (0.008)
10	Signal duty	0.4657	10.05 (0.070)	99.32 (0.009)	—

Table 5. The constant charge against the contention duty at common full-run access targets, sixteen held-out seeds. Paired differences use the delta method on matched seeds.

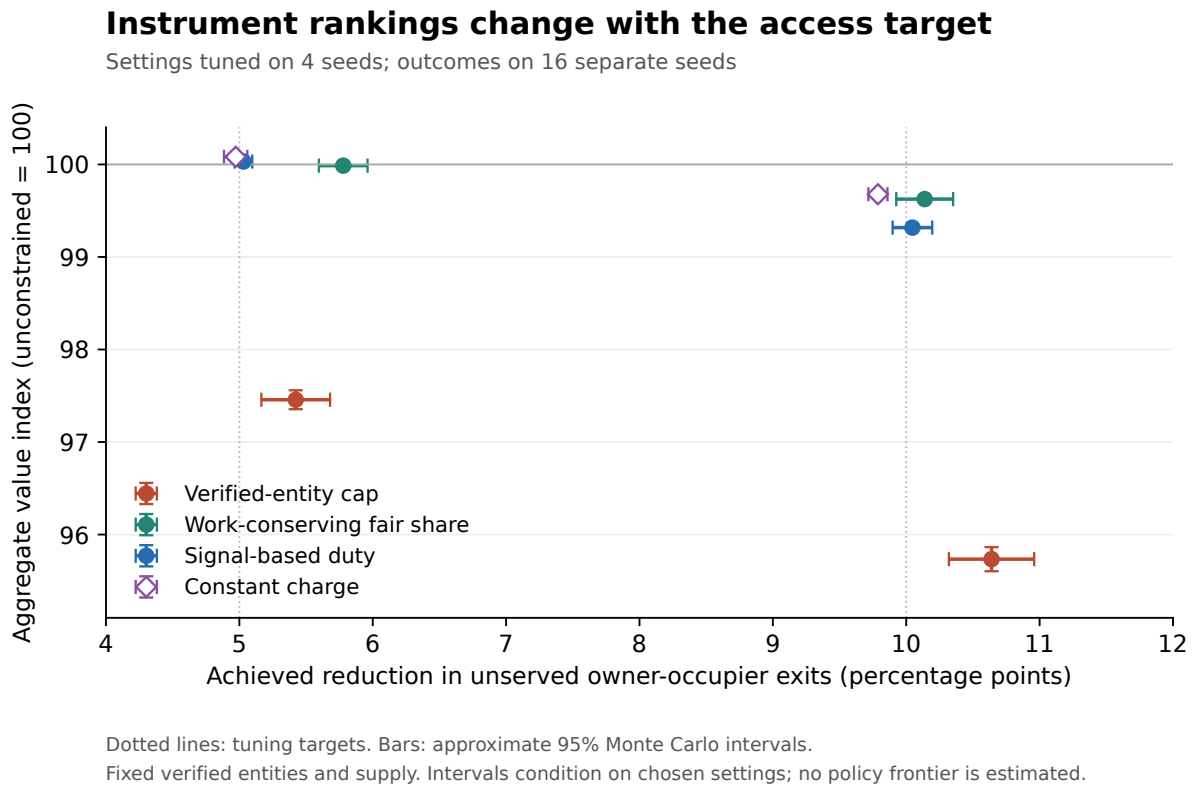


Figure 3. Settings tuned to common full-run access targets and evaluated on sixteen separate seeds, including the constant charge. Bars are approximate 95% Monte Carlo intervals conditional on the settings. Points are not joined into a frontier.

The constant charge, which ignores contention, keeps more value than the duty at both targets, by 0.05 and 0.36 index points (Table 5, Figure 3). Its access gains fall slightly short of the duty's, so the comparison is close but not exact.

The full-run starvation rate has a blind spot. Buyers still searching when the run ends count as neither served nor unserved, and instruments differ in how many they leave in that state. Following every owner-occupier who enters in periods 36 to 83 until they buy or leave by period 119 resolves the whole cohort.

Regime	Setting	Full-run gain (pp)	Cohort gain (pp; SE)	Cohort unserved (%)
Unconstrained	—	—	—	60.09
Static cap	336	5.42	13.53 (0.279)	46.56
Fair share	0.5000	5.78	7.91 (0.134)	52.18
Signal duty	0.2115	5.03	9.34 (0.072)	50.75
Constant charge	0.0779	4.97	7.15 (0.065)	52.94
Static cap	273	10.64	26.44 (0.318)	33.65
Fair share	0.4121	10.14	13.97 (0.164)	46.12
Signal duty	0.4657	10.05	18.04 (0.140)	42.05
Constant charge	0.1466	9.79	14.03 (0.091)	46.06

Table 6. Completed-cohort access at the settings of Table 5 and the common-target exercise, without retuning. The cohort averages 4,061 entrants per replication.

The full-run measure understates every instrument's effect on this cohort, by different amounts (Table 6). For the cap, a 5.42-point full-run gain is a 13.53-point cohort gain, 2.5 times larger; the fair share's 5.78 becomes 7.91, 1.4 times larger. The rankings also differ: at the five-point settings the fair share leads on full-run access and the duty leads on cohort access. The two measures describe different populations, and a comparison needs to state which one it targets.

Retuning to the cohort measure changes the settings and the comparison. Tuned on the original four seeds to five- and ten-point cohort gains and evaluated on sixteen new seeds, a five-point cohort gain needs a cap of 385 acquisitions per entity, not 336. At the ten-point cohort target the duty now keeps 0.032 more index points than the constant charge (paired SE 0.0033), while the constant charge delivers 1.607 points more full-run access (paired SE 0.044) and a cohort gain 0.026 points smaller (paired SE 0.044). Near five points the value difference, 0.0012 (paired SE 0.0017), is not resolved. The constant charge keeps more value when both instruments are tuned to full-run access, and the duty keeps more when both are tuned to cohort access. Neither result generalises beyond its stated population.

Waiting times carry the same trap. When costless entity splitting erodes a cap, the 99th-percentile wait of owner-occupiers who complete a purchase falls, even as starvation rises: the buyers who would have waited longest leave without buying and drop out of the sample. A waiting-time statistic for completed purchases means little without the share of entrants who never complete.

5.4 Which quota accounting

At low institutional participation the implemented fair share has no intermediate setting. With one institutional entity placing one bid per district-period, the unconstrained institutional share is 4.47%. Any positive period share admits the single institutional bid, and a zero share excludes it whenever an owner-occupier is waiting. The quota can only overshoot a 0.5- or 1-point access target or miss it entirely. The duty cannot reach those targets either within its allowed range.

Case	Target (pp)	Setting (share)	Access gain (pp; SE)	Value index (SE)
Baseline	5	0.5103	5.22 (0.095)	99.997 (0.007)
Baseline	10	0.4351	10.18 (0.103)	99.600 (0.015)
Low institutional capacity	0.5	0.0371	0.48 (0.016)	99.737 (0.008)
Low institutional capacity	1.0	0.0293	0.96 (0.023)	99.477 (0.011)

Table 7. The fair share with allowances that carry forward: credits accrue at the share times fresh listings each period and never expire. Share = 1 reproduces the held-out baselines, and share = 0 reproduces the original zero-share rule.

Replacing the period-reset allowance with credits that carry forward, one credit spent per institutional purchase while an owner-occupier waits, reaches gains of 0.48 and 0.96 points (Table 7). The original gap was a property

of the implementation, not of fair-share quotas. Unlimited banking is not proposed as a design: it changes the timing of institutional purchases, and limits on accumulation, expiry and borrowing are untested.

5.5 Robustness

The extended version reports the remaining sensitivity exercises; the main points are these. Letting 10% of accepted financed sales fail leaves the value indices and ranking similar and the fixed-benchmark break-even almost unchanged (11.8% against 11.7%), but it raises failed transactions under the duty by 21.6% and more than doubles them under the cap. Reduced search, with inspection probabilities of 0.5 and 0.25, preserves the ranking. Sweeping institutional buyers' mean valuation from 340,000 to 410,000 moves the duty's index by at most 0.06 points and raises the cap's cost from 8.6% to 16.0%. The baseline institutional share of 58% is far above observed markets; with the share lowered to 22.5% the duty and fair share barely act, reducing starvation by 0.7 and 0.1 points, while the cap still costs the most value at every share tested. The rate sweep finds the highest mean value index at $\tau_{\max} = 0.11$, not at the baseline 0.22.

$\tau_{\max}$	Starvation	OO share	Value index
0.00	41.9%	34.0%	100.00
0.11	39.2%	37.6%	100.12
0.22	36.6%	41.1%	100.02
0.33	34.3%	44.1%	99.77
0.44	32.3%	46.8%	99.42
0.60	29.8%	50.1%	98.79

Table 8. The duty's maximum rate, 8 replications, baseline offer discount of 12%. Access improves with the rate; the highest mean value index is at 0.11. At an offer discount of 5.6%, the ceilings 0.11, 0.22 and 0.44 all lose value.

6. The New York City evidence

The model's offer discount can be set against the companion estimates for New York City (Table 9). The comparison is a calibration check, not a test of any tax.

Market	Log-price gap, 100β	95% interval	Approx. cash discount
One- to three-family houses, headline	9.25	[7.17, 11.35]	8.8%
Houses, excluding same-surname deeds	4.87	[2.69, 6.76]	4.8%
Houses, cash buyer is a company	1.53	[-1.79, 5.44]	1.5%
Houses, cash buyer is another party	13.77	[10.96, 15.98]	12.9%
Houses, other party, excluding same-surname deeds	7.11	[4.48, 9.12]	6.9%
Condominiums	0.77	[-0.01, 1.56]	0.8%
Co-operatives	-1.42	[-3.70, 0.86]	-1.4%

Table 9. Financing-related price gaps, New York City 2016–2025, in log points; positive values mean higher adjusted prices for financed purchases. Cash discounts are $100[1 - \exp(-\beta)]$. Source: Loschi (2026a), version 3.4. House intervals are parcel-bootstrap intervals; the co-operative interval clusters by building. Property types differ in sample construction, so the rows are not a matched comparison across tenures.

Institutional share at $w = 12\%$	Interpolated break-even w	95% interval
22.4%	15.9%	[15.6, 16.1]
29.9%	14.2%	[13.8, 14.7]
40.2%	12.5%	[12.2, 12.7]
46.8%	12.1%	[11.8, 12.3]
52.8%	11.9%	[11.8, 12.1]
58.1%	11.7%	[11.5, 11.8]
63.5%	11.9%	[11.7, 12.0]
68.3%	11.8%	[11.7, 11.9]
73.7%	12.0%	[11.8, 12.1]
79.9%	12.3%	[12.2, 12.4]

Table 10. Offer discount at which the duty stops losing value on the fixed benchmark, across ten institutional-capacity settings, with the duty ceiling fixed at 0.22. Institutional shares are unconstrained shares at the baseline discount. Crossings are linearly interpolated between grid nodes two points apart, with eight replications per node. Intervals are 95% percentile intervals from 4,000 bootstrap resamples of the eight matched seeds (`threshold_intervals.py`); they reflect simulation noise, not error from the linear interpolation or the model.

Across the settings in Table 10 the break-even offer discount ranges from 11.7% [11.5, 11.8] to 15.9% [15.6, 16.1]. The only house group in Table 9 whose implied discount lies in that range is cash buyers other than companies, at 12.9%; excluding same-surname deeds reduces it to 6.9%. The company-cash discount is 1.5%, with an interval that includes zero. Transaction-price gaps do not separate a certainty premium from composition or completion costs. Mortgage denials, which precede accepted offers, cannot identify subsequent failure risk.

New York also charges purchase mortgages. Applying headline borrower-side rates gives averages of 1.41% of price for houses and 1.28% for condominiums before credits, exemptions and consolidation adjustments (Loschi 2026c). The model does not compare replacing that charge with taxing cash purchases.

6.1 The model at New York City values

Table 11 uses the company-cash gap of 1.53 log points to set the offer discount to 1.5%; its upper confidence endpoint gives 5.3%. These are sensitivity inputs. The price-gap interval includes zero. In the linked dataset, company cash buyers account for 19,019 of 166,227 house sales of at least \$100,000 with an ORDINARY linked deed, or 11.4%; all company buyers account for 22.8% (Loschi 2026d). Discrete capacity settings approximate those shares, producing 11.0% and 25.0% in the model.

At the narrower proxy, full-run starvation is 9.8%, against 41.9% at the paper baseline. The cap costs 0.77% of realised value per percentage point of full-run access, against 0.34% at baseline. The quota has no measured effect; the duty and constant charge each improve full-run access by about 0.2 points. Among entrants followed to purchase or exit, unserved rates are 14.9% without intervention, and the duty and constant charge improve access by 0.45 and 0.37 points. The constant charge does slightly better over the full run (0.24 against 0.20 points) and the duty slightly better in the cohort. Both differences are under 0.1 point, although each exceeds three paired Monte Carlo standard errors.

The constant charge equals the duty ceiling times its mean signal across district-periods. This matches an unweighted scheduled haircut, without equating charges on completed purchases or access. Table 11 holds instrument settings fixed; Section 5.3 remains the comparison at common access targets.

Small investors retain the convention $c_{SI} = \min(1, c_{OO} + 0.07)$, making them as certain as institutions at the company-gap inputs. Setting their certainty equal to owner-occupiers' reduces the duty's full-run gain from 0.20 to 0.18 points. With the offer discount set to zero, full-run starvation remains 9.5%, against 9.8% at the company-cash gap, and the duty still improves access by 0.18 points. At these inputs the certainty wedge explains little of the model's unserved exits, and the charges work mainly by limiting institutional purchases. In the upper-gap and headline-gap scenarios, the contemporaneous benchmark gives duty gains of 0.11 and 0.19 points, while the fixed oracle gives losses of 0.04 and 0.03 points.

Supply, arrivals, patience and valuations remain illustrative. Starvation is a model quantity, and the company shares concern a selected set of linked transactions. Company buyers include LLCs used by individuals; absence of a matched mortgage does not prove cash financing. Neither proxy identifies certain institutional buyers. The two headline inputs cannot establish New York City’s policy effects.

Scenario	Discount / model share (%)	Unservd: full / cohort (%)	Cap	Fair share	Duty	Constant
Company cash	1.5 / 11.0	9.8 / 14.9	-1.77 / -1.36	0.00 / 0.00	-0.20 / -0.05	-0.24 / -0.05
Upper gap	5.3 / 11.5	10.6 / 16.0	-2.15 / -1.47	0.00 / 0.00	-0.21 / -0.04	-0.26 / -0.04
All companies	1.5 / 25.0	18.8 / 27.2	-11.29 / -5.16	-0.46 / -0.06	-1.09 / -0.18	-1.21 / -0.17
Headline gap	8.8 / 12.1	11.1 / 16.8	-2.53 / -1.56	0.00 / 0.00	-0.20 / -0.03	-0.25 / -0.03
Paper baseline	12.0 / 58.1	41.9 / 59.8	-33.16 / -11.28	-17.14 / -1.78	-5.24 / +0.03	-4.85 / +0.10

Table 11. NYC proxy scenarios and the published baseline, twelve matched seeds. Each instrument cell is change in full-run starvation (percentage points) / realised value (percent), relative to its scenario’s unconstrained market; negative starvation means improved access. The unserved columns give unconstrained full-run and completed-cohort rates. The cohort enters in periods 36 through 83 and is fully resolved by period 119. Instruments retain the baseline settings; constant charges match the duty’s mean scheduled haircut. Shares are simulated purchase shares, not exact empirical targets. `nyc_anchored.json` supplies full-run results; `nyc_cohort_check.json` supplies cohort results, seed-level runs and paired Monte Carlo standard errors.

7. What the model leaves out

Bidding is not strategic. Valuations are private, fixed and bid truthfully, so the simulation tests what the instruments do to allocations, not how buyers would respond to them. A limited exercise lets institutions reallocate a fixed acquisition budget across districts in proportion to $(1 - \tau_{\max S_d})^\lambda$. For λ from zero to eight, the duty’s value index ranges from 99.60 to 100.22 and its starvation rate from 36.5% to 36.7%, so even the sign of its small aggregate effect is not stable once buyers move. Bid shading, entry and manipulation of the contention signal are not modelled. This is the largest gap.

The core results use one illustrative calibration; Section 6.1 varies two proxy inputs. Patience, the number of districts, valuation dispersion, the contention window and search friction leave the full ordering unchanged; arrival and listing rates do not. With owner-occupier arrivals doubled to 34 per district-period, which lowers the unconstrained institutional share to 38%, the fair share scores 100.20 and overtakes the duty at 100.09, while the duty still lowers starvation more (4.0 against 2.9 points). The cap is the most costly instrument at every point tested, at 97.3 in that case. Its place relative to the contention-linked instruments holds across the reported exercises; the ordering within that pair does not. The baseline institutional share of 58% is also far above observed markets, where institutional purchases run at a few per cent nationally.

The value index is not welfare. It counts realised willingness to pay and leaves out renters, the use of any revenue, administrative costs and differences in the marginal utility of income. The supply exercise lets listings respond to seller proceeds; it has no construction sector, rents or entry. The horizon is finite, so the full-run starvation rate is not a lifetime purchase probability, and entity splitting is costless, so the cap’s decay is an upper bound on avoidance without enforcement.

8. Implementation

Each instrument needs information the model takes for granted.

The duty needs an exit registry. Its signal counts owner-occupiers who leave without buying. Recording that requires a register that observes entry into the market and distinguishes purchase, withdrawal and a move to another market; routine transaction records do not show it. The signal can be pushed in either direction. Institutions could depress it by withdrawing bids during the measurement window. Anyone who gains from a higher duty could inflate it with entries or withdrawals staged to register as unserved exits. A workable version would count only verified participation, such as a mortgage pre-approval or a deposit, keep the records with a party that has no stake in the rate, and limit how far the rate can move in one period. The constant charge

needs none of this, which is the practical side of Section 5.3: the duty's value advantage is small, depends on the access population, and has to be weighed against the cost of measuring contention.

The cap needs ownership aggregation. A per-entity limit binds a legal person, and in the model a single owner who can form entities at no cost defeats it with 25 of them. An enforceable cap would need a definition of common control, disclosure of beneficial owners, verification and sanctions. Federal beneficial-ownership reporting has exempted domestic companies since 2025, so a state rule would have to supply that information itself. A charge on each purchase avoids the gain from multiplying entities, although it still needs an enforceable definition of who pays.

The fair share's release rarely binds. The quota is work-conserving: it lifts when no owner-occupier is waiting. At the baseline's level of excess demand someone is always waiting, so the simulation tests the quota and not its release. Its behaviour in slack markets is untested.

Classification decides who is reached. Payment method, institutional status and owner-occupation are different categories. Some owner-occupiers pay cash and some investors borrow. The model charges institutions. The 2026 New York proposal would have charged cash purchases of \$1 million or more, and a payment-method test can be avoided with a short-term loan repaid after closing. A charge on all non-owner-occupier purchases would also reach small investors and would need its own simulation.

Coverage and geography matter. A tax reaches only what its statute defines. New York's recording tax applies to mortgages on real property, so co-operative share loans fall outside it, while the mansion tax expressly includes co-operative units. Buyers can also move purchases to places the policy does not cover; the district exercise in Section 7 captures only a limited version of that response.

9. Conclusion

In this model, instrument rankings depend on the efficiency benchmark and the access population. A benchmark computed over waiting buyers moves with the policy and can reverse the sign of the duty's effect. Following entrants to purchase or exit changes measured access gains and can reverse their ordering. The constant charge and contention duty also swap places when retuned to different access populations. Carrying quota allowances forward resolves a limit created by resetting them each period.

At the baseline, the duty delivers modest access gains with little change in realised value, while the cap delivers the largest gain at the largest loss. Buyer-class weights above about 1.66 favour the cap among the three original instruments; entity splitting and elastic supply weaken it. At the company-cash proxy values, the duty lowers full-run unserved exits by 0.2 points, while the cap lowers them by 1.8 points at a 1.4% loss in realised value. Removing the cash discount there lowers unserved exits only from 9.8% to 9.5%, so at those inputs certainty explains little of the access shortfall.

The New York City inputs leave the model's other market conditions illustrative. Transaction composition affects the observed price gap, and company status does not identify certainty or beneficial ownership. Policy evaluation needs a fixed efficiency benchmark and an explicit access population. Waiting-time reports should include exits without purchase, and quota comparisons should specify how allowances accumulate.

Declarations

Declaration of generative AI use. During preparation the author used Anthropic's Claude for drafting, coding and literature assistance, including the preparation of this shortened version, and OpenAI's Codex for review, reproduction checks and figure revision. For version 6.1, Claude ran the New York City sensitivity exercise, and Codex checked it and ran the completed-cohort diagnostic. Computational reviews reported with earlier versions were performed by separate AI agents, not by human referees. The author is responsible for the content.

Competing interests. The author founded HolaCasaClara, a residential real estate service in Latin America with no operations or interests in the jurisdictions discussed here. No party funded, reviewed or approved the manuscript.

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Data availability. The package retains the extended version’s model and results (Loschi 2026b) and adds the scripts and outputs for Section 6.1. The version family is <https://doi.org/10.5281/zenodo.22164073>. New York City inputs come from the companion study (Loschi 2026a) and linked dataset (Loschi 2026d). Their source hashes and the share calculation accompany this version.

Prior dissemination. Earlier versions are posted on Zenodo under the concept DOI above and on SSRN.

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Appendix A. Reproducibility

The package retains the extended version’s simulation code and adds the NYC sensitivity scripts for Section 6.1. `clean_reproduce.py` regenerates the baseline, the offer-discount and rate sweeps, the supply and entity experiments, the capacity and valuation sweeps, the break-even schedule of Table 10 and the class-weight table. The main scripts are `replication/sim_v6.py` (baseline and failed-transaction variants), `replication/sim_v6_supply.py` (listing response), `replication/sim_v6_search_fixed.py` (reduced search), `replication/displacement.py` (Section 4.2), `replication/welfarist.py` (Table 2), `replication/breakeven_schedule.py`

(Table 10), `equal_access.py` (common targets), `constant_charge.py` (Table 5), `check_completed_cohorts.py` (Table 6), `cohort_targets.py` (cohort retuning) and `carry_share.py` (Table 7). `threshold_intervals.py`, new in version 6.0, reruns seed by seed the grid nodes around each break-even crossing, checks that their means reproduce the published schedule exactly, and bootstraps the matched seeds for the intervals in Section 5.2 and Table 10.

The baseline uses seed 20260824 and twelve replications, advancing the seed by 100 between runs. The common-target exercises tune on seeds $22260824 + 100i$ for $i = 0, \dots, 3$ and evaluate on $23260824 + 100i$ for $i = 0, \dots, 15$; the cohort retuning is evaluated on sixteen further seeds disjoint from both. Within a seed every policy sees identical arrivals, valuations and listings, and the oracle denominators match; the reduced-search runs draw search from a separate generator so that this also holds there.

On 21 September 2026 the package was extracted into a fresh directory with reads from the original workspace blocked, and every result was regenerated, including the 1,020 runs behind the constant-charge, cohort and carry-forward experiments. All matched the shipped outputs to floating-point tolerance. The New York City estimates in Table 9 come from the companion study’s version 3.4 (Loschi 2026a). For Table 11 run `replication/nyc_anchored.py`, then `replication/nyc_cohort_check.py`. The original model is unchanged; the cohort diagnostic reproduces every original summary exactly. `replication/verify_nyc_inputs.py` recomputes the share proxies from the separately deposited dataset; instructions and input hashes are in the README and `nyc_inputs.json`.